



A prospective audit of regional anaesthesia failure in 5080 Caesarean sections★

S. M. Kinsella

Consultant Anaesthetist, Department of Anaesthesia, St Michael's Hospital, Bristol BS2 8EG, UK

Summary

Anaesthesia for Caesarean section was audited over a 5 year period: 5080 cases were performed using spinal 63%, epidural top-up 26%, combined spinal-epidural 5% and primary general anaesthesia 5%. The rate of general anaesthesia conversion of regional anaesthesia was 0.8% for elective and 4.9% for emergency Caesarean section compared to Royal College of Anaesthetists targets of 1% and 3%. The rate of conversion of regional to general anaesthesia in category 1 Caesarean section was 20%. A total of 8% of women had general anaesthesia when both primary general and conversion of regional anaesthesia were combined. The rate of failure to achieve a pain-free operation was 6% with spinals, 24% with epidural top-up and 18% with combined spinal-epidural. Besides the type of anaesthesia and operative urgency, other factors associated with pre-operative failure of regional anaesthesia included body mass index, no previous Caesareans, and indication for Caesarean of acute fetal distress or maternal medical condition. Inadequacy of pre-operative anaesthetic block and duration of surgery were important risk factors for intra-operative failure. For spinal anaesthesia, use of a spinal opioid was associated with less pre-operative failure. For epidural top-up anaesthesia, lower epidural top-up volume was associated with less pre-operative failure, and use of adrenaline was associated with both less pre-operative and intra-operative failure. The rate of serious adverse incidents was 1 : 126 with general anaesthesia and 1 : 501 with regional anaesthesia.

Correspondence to: Stephen Michael Kinsella

E-mail: stephen.kinsella@ubhlt.nhs.uk

★Data presented in part at the Obstetric Anaesthetists' Association annual scientific meeting, London 2005.

Accepted: 21 January 2008

Regional anaesthesia is the predominant method of anaesthesia for Caesarean section in the UK. Although driven by concerns about the safety of general anaesthesia highlighted in the national Confidential Enquiries into Maternal Mortality, its popularity has been ensured by a high degree of reliability. In contrast to non-obstetric regional anaesthesia, systemic sedative and analgesic adjuncts are not used routinely because of concerns about maternal pulmonary aspiration and fetal effects.

The percentage use of general anaesthesia for Caesarean section has become a marker of the quality of the obstetric anaesthetic service [1–3]. As the overall use of general anaesthesia has declined, urgency of Caesarean section and failure of regional anaesthesia have become the main indications [3, 4]. A four category classification of Caesarean section urgency was described in 2000 with the aim of improving audit data collection in comparison to the binary elective/emergency classification [5].

Failure of regional anaesthesia is defined in several ways. Objective outcomes include conversion to general anaesthesia [4, 6, 7], conversion to any different form of anaesthesia [4, 8], or pain during surgery [9, 10]. In research studies of epidural top-up solutions where onset time is the primary end-point, failure may be defined as the inability to achieve a defined degree of nerve block suitable for Caesarean section [11, 12]. These studies have not established an ideal top-up solution. Research studies on patient position during establishment of spinal anaesthesia have shown differences in onset time but have been too small to allow analysis of failure rate [13].

This audit was planned when there were few published figures on the failure rate of contemporary obstetric regional anaesthesia, that is with a high rate of spinal or combined spinal-epidural (CSE) anaesthesia and with frequent use of spinal and epidural opioids. The primary aim was to allow the setting of standards for regional

anaesthesia failure for patient information and benchmarking. The secondary aim was to investigate the influence of urgency and anaesthetic management on failure rate.

Method

The audit was registered with the hospital Clinical Audit Committee, and the local Research Ethics Committee was informed. Data were collected on Caesarean sections carried out at St. Michael's Hospital, Bristol from 1st June 1999 to 31st May 2004. The primary anaesthetic was defined as a regional anaesthetic if a needle was inserted into the back, even if no further progress was achieved in locating the correct space. The datasheet included a flow diagram which requested details of problems at successive stages of the establishment and management of the anaesthetic (Appendix). The definition of a satisfactory regional anaesthetic block for Caesarean section was the loss of cold sensation, using ethyl chloride spray, from T4 (the nipples) down to S5 (the buttocks), as well as anaesthesia (no feeling) to a 19 gauge needle inserted at several points along the line of surgical incision at T12. Pain during the operation was split into mild or severe, based on a cut-off for mild pain of up to 1 mg alfentanil or 100 µg fentanyl, and the use of only one type of analgesia.

Initially, data were collected only on Caesarean sections under regional anaesthesia. Starting on 1st June 2000 (Year 2), all Caesarean sections that had primary general anaesthesia were also logged using a separate audit numbering sequence. The indication for the use of general anaesthesia was included on the data form.

The urgency of Caesarean section was classified using a 5-point scale in Year 1 and a 4-point scale from Year 2 [5]. The urgency grade in Year 1 cases has been converted to the equivalent 4-point scale by combining the original grades 2 and 3 [5].

Spinal anaesthesia was the preferred method and the use of the right lateral position for insertion was encouraged. The standard dose was 2.5 ml hyperbaric bupivacaine 0.5% with 300 µg diamorphine. Spinal fentanyl was occasionally used, especially for more urgent cases. When an epidural was used for analgesia during labour, it was usually topped-up for Caesarean section. The drugs used to top up were not standardised. These were bupivacaine and lidocaine, to which might be added adrenaline 5 µg.ml⁻¹ (1 : 200 000), sodium bicarbonate 8.4%, fentanyl or diamorphine. Laevo-bupivacaine became available towards the end of the audit period. CSE and microspinal catheters were reserved for complex cases, such as those that required slow onset of the regional block, women with anatomical difficulties or when prolonged surgery was anticipated. General anaesthesia was discouraged unless specifically indicated. If accidental

dural puncture occurred during labour, the epidural catheter was sometimes inserted into the spinal space in order to use it for labour analgesia.

Statistical analysis was performed to investigate factors associated with pre-operative or intra-operative failure. Pre-operative failure was defined as conversion to another anaesthetic before the operation started or failure to achieve a satisfactory block (as defined above). Intra-operative failure was defined as unsatisfactory anaesthesia that required analgesia [10]. This excluded cases where conversion to another anaesthetic had been performed before surgery.

The analysis was carried out initially on all regional anaesthesia cases. Univariate analysis of factors that might influence pre-operative failure was performed first using chi-squared tests. Factors from the audit database included audit year, type of anaesthesia, labour epidural in place but not topped up, indication for Caesarean, whether the woman was in labour and Caesarean urgency. Data from the labour ward obstetric database included height, weight at booking, body mass index (BMI), gestation, parity, number of previous Caesarean sections, neonatal weight and decision-delivery interval. Forwards and backwards stepwise logistic regression analysis was then used to determine which factors were significant predictors of the outcome. A *p* value < 0.05 was used to define statistical significance. The same process was repeated for intra-operative failure, but including the factors 'block adequate pre-operatively' and duration of surgery.

The same analysis was performed on the sub-groups of women with spinal and epidural top-up anaesthesia. In the spinal group, factors specific to this type of anaesthesia were also included; these were insertion position, indication for position if sitting (change from lateral, obese, preference of the anaesthetist, other/unknown), position after insertion, local anaesthetic volume and use of spinal opioid. In the epidural top-up group, the specific anaesthetic factors included were type of local anaesthetic (bupivacaine, lidocaine, 50 : 50 bupivacaine:lidocaine, and 'other' including laevo-bupivacaine and non-standard local anaesthetic mixtures), volume of local anaesthetic, use of adrenaline, use of bicarbonate and dose of epidural opioid (including zero dose).

Results

During the period from 1 June 1999 to 31 May 2004 the annual number of deliveries ranged between 4329 and 4773 and the Caesarean section rate between 22.0% and 24.2%. There was a gradual increase in the annual number of Caesarean sections from 922 to 1142, which was accompanied by an increase in spinals from 541 in Year 1 to 751 in Year 5. The number of primary general anaesthetics showed a tendency for a gradual

decline, the highest being 81 in Year 2 and the lowest 36 in Year 4.

A total of 5080 Caesarean sections were performed during the audit, of which 4806 had regional anaesthesia. There were 3224 spinals, 1286 top-up epidurals (including a small number of CSE inserted during labour), 217 standard dose CSEs, 55 low-dose CSEs (< 2.5 ml bupivacaine 0.5%), 11 microspinal catheters (Kendall Co-span 28 G through 22 G needle), 11 primary epidurals, and two spinal catheters following accidental dural puncture during attempted labour epidural. The process of establishment and management of the three main types of regional anaesthesia is illustrated in flow diagrams (Figures 1–3). In one spinal anaesthetic, severe pain occurred after delivery, general anaesthesia was

refused and an epidural was sited and provided satisfactory anaesthesia; this case is counted as a general anaesthetic conversion. One epidural top-up with pre-operative high block requiring intubation is also counted as a general anaesthetic conversion.

Two microspinal catheters were converted to epidural anaesthesia and one had general anaesthesia for intra-operative pain. Three primary epidurals needed analgesic supplements, of whom two had a satisfactory block before surgery.

Table 1 shows the indications for Caesarean section and the indications for primary general anaesthesia. Fetal indication was not usually noted as an indication for general anaesthesia. However, if urgency was specified as the indication for general anaesthesia, the indication for

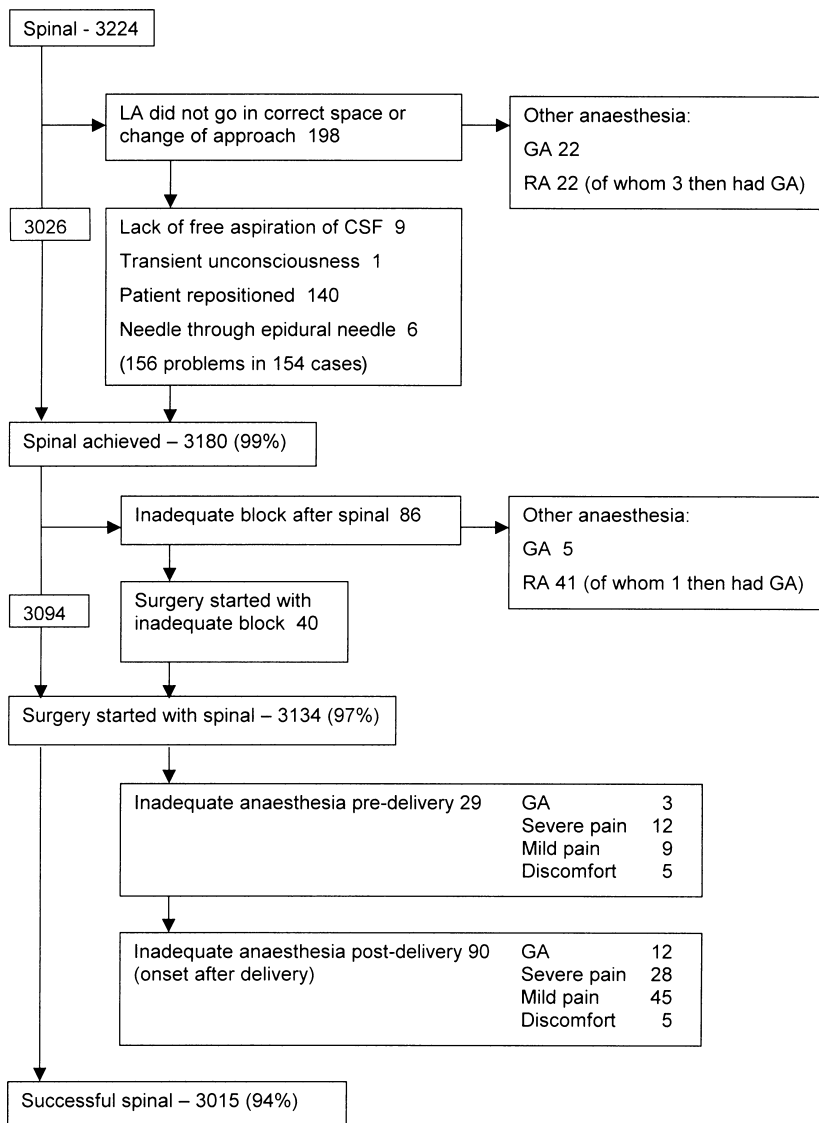


Figure 1 Flow diagram of spinal management. A successful spinal is one that was used for Caesarean section with no intra-operative discomfort. [LA, local anaesthetic; GA, general anaesthesia, RA, regional anaesthesia, for definition of severe and mild pain see text].

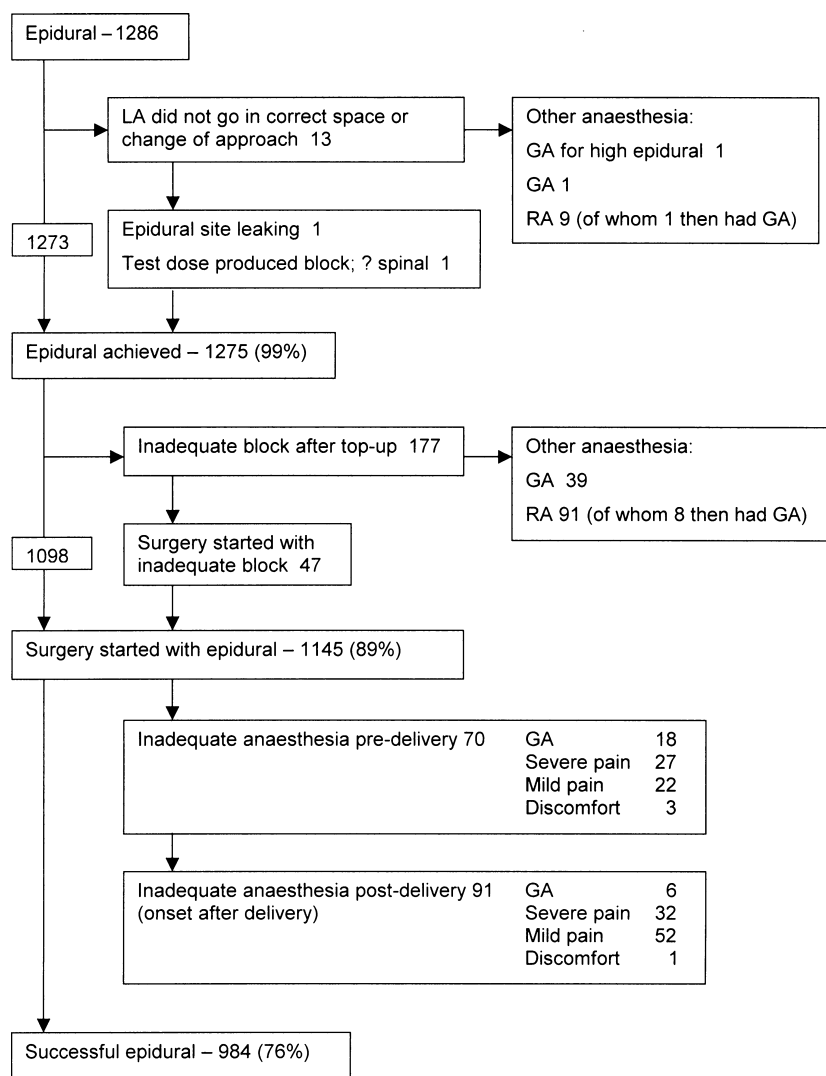


Figure 2 Flow diagram of top-up epidural management. A successful epidural is one that was used for Caesarean section with no intra-operative discomfort. [LA, local anaesthetic; GA, general anaesthesia; RA, regional anaesthesia, for definition of severe and mild pain see text].

the Caesarean section was usually fetal distress. Table 2 shows patterns of anaesthetic provision and conversion of regional anaesthesia to general anaesthesia depending on the urgency of Caesarean section. The rate of general anaesthetic conversion for spinals was 8% in category 1 compared to 0.5% in category 4, and for epidurals the rate was 25% and 2.4% in categories 1 and 3 respectively. General anaesthetic conversion was pre-operative in 83% category 1 spinals and 90% category 1 top-up epidurals, in contrast to pre-operative conversion in 68% of all spinals and 68% of all top-up epidurals.

Most of the women having a category 1 operation (83%) and category 1–3 operation (75%) were in labour.

There were 106 cases in which an epidural had been established during labour but was not topped up for Caesarean section. The majority of these were not topped up because of poor quality block, but other

reasons included epidural catheter misplaced or disconnected, epidural only recently inserted, lack of time for top up, maternal anxiety and previous accidental dural puncture. Of these 106 patients, 86 had a primary regional block for Caesarean section and 20 had general anaesthesia: 16 for urgency and four for maternal preference.

Two cases with an epidural catheter placed spinally after accidental dural puncture had unreliable analgesia during labour and there was no attempt to top up the 'intrathecal catheter'. Primary spinal anaesthesia was planned, but in one case the spinal could not be located and general anaesthesia was given.

Table 3 shows the number of women having intra-operative pain and conversion of regional anaesthesia to general anaesthesia related to the adequacy of block before surgery.

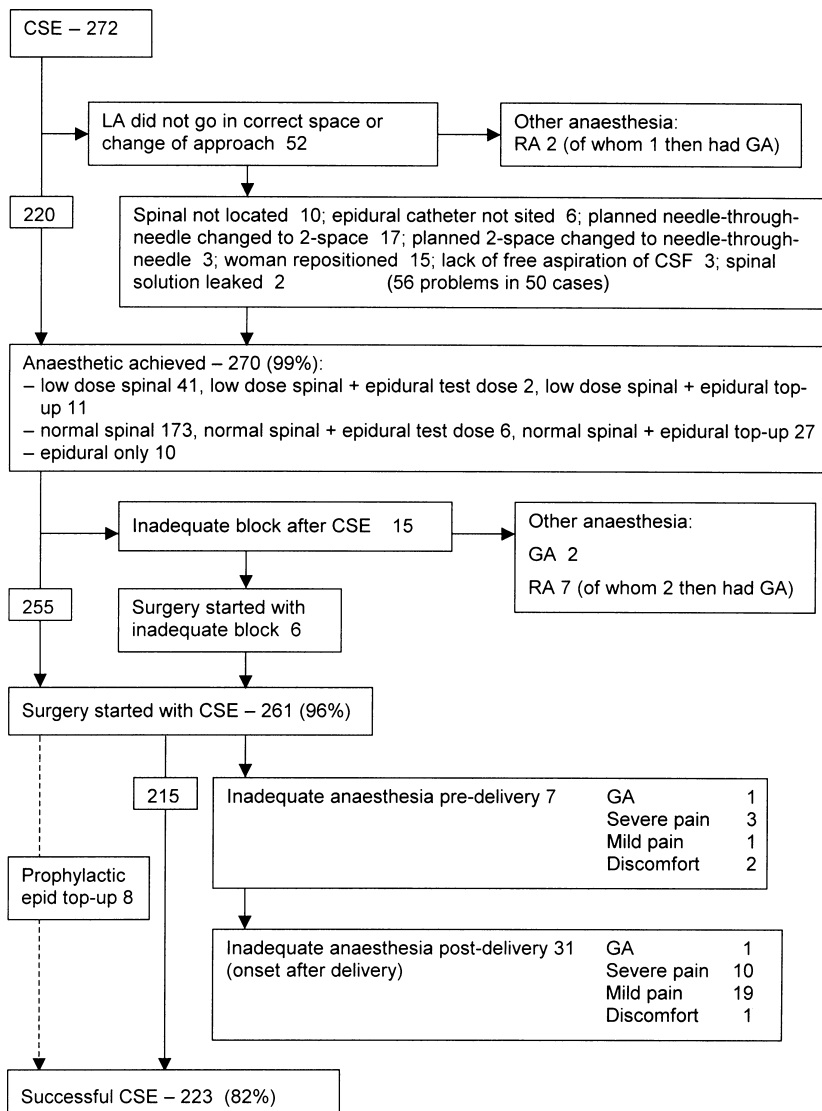


Figure 3 Flow diagram of combined spinal-epidural (CSE) management. A successful CSE is one that was used for Caesarean section with no intra-operative discomfort. [LA, local anaesthetic; GA, general anaesthesia; RA, regional anaesthesia, for definition of severe and mild pain see text].

Critical incidents

A box asking for reporting of critical incidents was added to the data collection form from regional anaesthesia case no 1300 (general anaesthesia case number 44). These are presented in decreasing order of seriousness:

Primary general anaesthesia

- Category 1; initial attempt at intubation failed, mask ventilated before further successful attempt.

Primary general anaesthesia with labour epidural established but not topped up

- (*regional anaesthesia problem*) category 3; intravenous epidural catheter; epidural ineffective and top-ups 'go to my head'.

Conversion of regional anaesthesia to general anaesthesia

- Category 3; epidural top-up for failure to progress in labour in a pre-eclamptic woman; pain on entering the peritoneum, general anaesthesia conversion followed by bronchospasm and pulmonary oedema requiring ITU.
- Category 2; acute fetal distress after eclampsia, CSE attempt produced inadequate block, pre-operative general anaesthesia was followed by persisting hypoxaemia and hypotension, transferred to ITU.
- Category 2; maternal distress about sensation of spinal block, pre-operative general anaesthesia conversion and failed intubation at first attempt.
- Category 4; failure to site spinal; general anaesthesia conversion, endobronchial intubation with desaturation, bradycardia and hypotension.

Table 1 Indications for Caesarean section and indications for primary general anaesthesia. These totals are > 100% as multiple indications could be assigned.

Indications for CS	Previous Caesarean/ to avoid vaginal delivery	Failure to progress in labour/failed instrumental delivery	Acute fetal distress/ cord prolapse	Abnormal fetal position/multiple gestation	Maternal medical/ other	Chronic fetal/ preterm	Haemorrhage/ placenta praevia
CS under RA	33%	27%	22%	22%	13%	7%	4%
CS under GA	17%	1%	51%	9%	22%	8%	39%
Category 1 CS (years 2–5)	2%	8%	86%	8%	6%	33%	23%

Indications for GA	Urgency	Maternal choice	Regional anaesthesia contraindicated	Fetal indication	Hypovolaemia	Other
All CS	51%	35%	16%	13%*	10%	9%
Category 1 CS (years 2–5)	86%	10%	12%	23%**	14%	3%

*51% of GA had fetal indication if 'Indication for CS, acute fetal distress/cord prolapse' is included.

**83% of category 1 GA had fetal indication if 'Indication for CS, acute fetal distress/cord prolapse' is included.

CS, Caesarean section; RA, regional anaesthesia; GA, general anaesthesia.

Table 2 Caesarean section urgency, type of primary anaesthesia and conversion of regional anaesthesia to general anaesthesia during years 2–5.

	Urgency				Total
	1	2	3	4	
Total CS	324 (8%)	1045 (25%)	1180 (28%)	1608 (39%)	4158
Primary RA	209 (5%)	1020 (26%)	1155 (29%)	1571 (40%)	3955
Primary GA	115 (57%)	26 (13%)	25 (12%)	37 (18%)	203
GA conversion of RA	29 (26%)	48 (44%)	21 (19%)	12 (11%)	110
Total GA	144 (46%)	74 (24%)	46 (15%)	49 (16%)	313
Spinal	135 (5%)	489 (18%)	651 (24%)	1408 (52%)	2683
GA conversion	11	11	10	7	39
Epidural top-up	72 (7%)	505 (49%)	452 (44%)	0	1029
GA conversion	18	35	11		64
CSE–normal dose	0	14 (7%)	42 (22%)	127 (70%)	183
GA conversion	0	2	0	3	5
CSE–low dose	0	8 (20%)	7 (17%)	26 (63%)	41
GA conversion	0	0	0	1	1
Microspinal catheter	0	1	2	8	11
GA conversion	0			1	1
Primary epidural (no GA conversions)	2	1	1	2	6
Spinal catheter (epidural) (no GA conversions)	0	1	1	0	2

Data are number (% of total in row); number of conversions to general anaesthesia in italics. CS, Caesarean section; RA, regional anaesthesia; GA, general anaesthesia; CSE, combined spinal-epidural.

- (*regional anaesthesia problem*) category 4; inadequate spinal required general anaesthesia conversion before delivery, asystole during peritoneal stretching.

Regional anaesthesia

- Category 3; epidural topped up for a trial of instrumental delivery with 20 ml bupivacaine 0.5% produced C2 block with respiratory inadequacy requiring intubation (counted as general anaesthesia conversion in totals).
- Category 3; accidental dural puncture on epidural insertion during labour managed as spinal catheter but block poor; spinal inserted for Caesarean section with 2.5 ml hyperbaric bupivacaine 0.5% and 200 µg diamorphine produced C5 block and transient loss of consciousness; she was mask ventilated but recovered consciousness before intubation could be performed.
- Category 1; fetal distress, tachycardia noted just prior to epidural top-up, pulmonary oedema developed and supraventricular tachycardia diagnosed, general

Table 3 Incidence of pain and intra-operative conversion to general anaesthesia depending on adequacy of block before surgery in 4540 cases using regional anaesthesia.

	No pain	Pain	GA
Total			
Block adequate (4439)	4171	268	32
Block inadequate (101)	66	35	9
Spinal			
Block adequate (3094)	2993	101	14
Block inadequate (40)	30	10	1
Top-up epidural			
Block adequate (1090)	958	132	16
Block inadequate (55)	30	25	8
CSE			
Block adequate (255)	220	35	2
Block inadequate (6)	6	0	0

Cases where there was pre-operative conversion to another regional anaesthetic have been excluded.

GA, general anaesthesia; CSE, combined spinal-epidural.

anaesthesia induced for cardiorespiratory control and rapid delivery.

- Category 2; accidental dural puncture suspected after insertion of an epidural during labour because of high block; when Caesarean section was required for fetal distress the epidural was topped up with bupivacaine 20 ml and fentanyl 100 µg; decreased consciousness ensued and general anaesthesia was induced.
- Category 2; fetal distress, standard spinal insertion, patient developed severe pain after delivery but refused general anaesthesia, agreed to epidural which was performed in the lateral position, once established surgery proceeded painlessly (counted as general anaesthesia conversion in totals).
- Category 4; pulmonary oedema after delivery in a woman with severe aortic stenosis having low dose CSE.
- Category 4; spinal, asystole following skin incision; several years before she had an almost identical episode during her first Caesarean section.
- Category 2; spinal misplacement of epidural dose; a 3 ml test dose of 0.5% bupivacaine and 1 : 200 000 adrenaline with 7.5 µg fentanyl produced a C5 block with arm weakness.
- Category 3; epidural top-up for failure to progress, transient drowsiness and unresponsiveness during surgery responded to naloxone.
- Category 3; blood aspirated from epidural catheter after 5 ml of top-up had been given; spinal inserted for operation.

In addition to these cases, there were six high blocks with respiratory problems, three high blocks with arm weakness, six high blocks, three bradycardias, 10 hypotensions, three supraventricular tachycardias,

five non-anaesthetic drug or latex reactions and three miscellaneous.

During this period, there were 159 primary general anaesthetics and 93 regional anaesthetics converted to general anaesthesia. There were five critical incidents (two serious) associated with general anaesthesia, all but one in cases of general anaesthetic conversion. The rate of incidents was 2%. There were 51 incidents (seven serious) related to the regional anaesthesia in 3506 regional anaesthetics, a rate of 1.4%.

During the period that critical incidents were recorded, 68 cases had a spinal anaesthetic in the presence of an epidural that had not been topped up for Caesarean section. Mean volume of hyperbaric bupivacaine 0.5% was 2.4 ml [range 1.5–2.5 ml]. One high sensory block (T1) was noted. In 71 cases with an epidural that was apparently in the correct space but failed to produce an adequate block for Caesarean after being topped-up, a spinal anaesthetic was given. Mean volume of hyperbaric bupivacaine was 1.6 ml [range 1.0–2.5 ml]. Seven high sensory blocks were noted, ranging from C3 to C8 where documented. Five of these were accompanied by dyspnoea and one by hand weakness. In three of the 71 cases, the spinal injection was apparently made into the correct space but did not produce any change in block, possibly because it was made into the epidural space containing local anaesthetic rather than the subarachnoid space.

Statistical analysis

Using univariate analysis, there was no effect on failure rate of audit year and epidural drugs including local anaesthetic, opioid or bicarbonate. There was a significant effect of epidural adrenaline on the risk of both pre-operative and intra-operative failure, and spinal insertion position on the risk of pre-operative failure. Further statistical results are available from the author. The results of the logistic regression analyses are shown in Tables 4–6.

Discussion

This audit observed the anaesthetic practice in a large teaching hospital. The overall use of anaesthetic techniques in the 5-year period (general anaesthesia 5%, spinal 63%, epidural top-up 26%, CSE 5%), is not dissimilar to national data from 2004 (general anaesthesia 11%, spinal 55%, epidural 20%, CSE 10%) [14], although the general anaesthesia rates are not strictly comparable as the national data may include cases of failed regional block.

National standards for rate of regional anaesthesia and conversion of regional anaesthesia to general anaesthesia were published after the start of this audit. Our data are largely compliant with standards first set by the Royal

Table 4 Logistic regression analysis of factors associated with regional anaesthesia failure. Pre-operative failure includes conversion to another anaesthetic or failure to achieve a satisfactory block; intra-operative failure includes unsatisfactory anaesthesia that required treatment.

Pre-operative	Odds ratio	95% CI	Chi-square	p value	Intra-operative	Odds ratio	95% CI	Chi-square	p value
Type of anaesthesia					Type of anaesthesia				
Epidural top-up	3.15	[2.39–4.14]	67.4	<0.001	Epidural top-up	5.73	[4.30–7.65]	144.6	<0.001
Spinal	1.00				Spinal	1.00			
CSE	1.60	[0.90–2.82]			CSE	3.04	[1.99–4.65]		
Indication for CS – acute fetal distress (including cord prolapse)					Preexisting labour epidural not used for CS				
No	1.00		6.6	0.01	No	1.00		6.7	0.01
Yes	1.63	[1.12–2.37]			Yes	3.01	[1.30–6.96]		
Indication for CS – maternal medical					Indication for CS – twins				
No	1.00		8.3	0.004	No	1.00		5.1	0.024
Yes	1.81	[1.21–2.72]			Yes	0.38	[0.17–0.88]		
CS urgency category					Indication for CS – failed instrumental delivery				
1	2.45	[1.36–4.40]	12.5	0.006	No	1.00		8.0	0.005
2	1.27	[0.79–2.04]			Yes	0.39	[0.20–0.74]		
3	1.29	[0.87–1.90]			Block adequate pre-operatively				
4	1.00				No	6.58	[4.13–10.50]	62.5	<0.001
BMI					Yes	1.00			
≤25	0.80	[0.60–1.06]	11.7	0.039	Baby's weight; g				
25.1–30	1.00				≤3000	1.37	[0.96–1.94]	13.1	0.004
30.1–35	0.71	[0.46–1.09]			3001–3400	0.93	[0.66–1.32]		
35.1–40	0.95	[0.55–1.63]			3401–3800	1.00			
>40	1.76	[0.96–3.21]			>3800	0.68	[0.48–0.98]		
Unknown	0.69	[0.47–1.02]			Duration of operation; min				
Previous CS					0–60	1.00		96.3	<0.001
0	1.00		7.5	0.024	61–90	1.50	[0.93–2.40]		
1	0.60	[0.41–0.86]			>90	6.88	[4.25–11.15]		
≥2	0.83	[0.44–1.57]			Unknown	2.23	[1.42–3.50]		

Table 5 Logistic regression analysis of factors associated with spinal failure. Pre-operative failure includes conversion to another anaesthetic or failure to achieve a satisfactory block; intra-operative failure includes unsatisfactory anaesthesia that required treatment.

Pre-operative	Odds ratio	95% CI	Chi-square	p value	Intra-operative	Odds ratio	95% CI	Chi-square	p value
Opioid					Duration of operation (min)				
No	5.51	[1.99–15.27]	7.8	0.005	0–60	1.00		35.7	<0.001
Yes	1.00				61–90	2.41	[0.81–7.16]		
Indication for Caesarean – acute fetal distress (including cord prolapse)					>90	9.10	[3.12–26.56]		
No	1.00		5.4	0.020	Unknown	2.93	[1.03–8.35]		
Yes	1.78	[1.12–2.84]			Block adequate pre-operatively			26.5	<0.001
Indication for Caesarean – preterm					No	11.71	[5.36–25.6]		
No	1.00		10.3	0.001	Yes	1.00			
Yes	3.09	[1.67–5.71]							
Parity									
0	1.00		6.7	0.035					
1	0.59	[0.37–0.95]							
2+	0.56	[0.31–1.00]							

College of Anaesthetists (RCA) in 2000 and reiterated in 2006 [3]. The regional anaesthesia rate for elective (category 4) Caesarean section was 98%, for emergency (category 1–3) Caesarean section was 93%, and for category 1 Caesarean section was 64%, compared to targets of 95%, 85% and 54% respectively.

The rate of conversion of regional anaesthesia to general anaesthesia was 0.8% for elective Caesarean section compared to a target of 1% [3]. The rate of

conversion of emergency Caesarean section depends on a particular aspect of the RCA definition. Where a general anaesthetic is given in the presence of a labour epidural which is not topped up, this is counted as a converted regional rather than a primary general anaesthetic. Using this definition, our rate for general anaesthesia conversion was 4.9% compared to the RCA target of 3%, but was 4.1% if these cases were not included as converted regionals. There is no published data on the expected

Table 6 Logistic regression analysis of factors associated with epidural failure. Pre-operative failure includes conversion to another anaesthetic or failure to achieve a satisfactory block; intra-operative failure includes unsatisfactory anaesthesia that required treatment.

Pre-operative	Odds ratio	95% CI	Chi-square	p value	Intra-operative	Odds ratio	95% CI	Chi-square	p value
Local anaesthetic volume; ml					Use of adrenaline				
0–14	0.75	[0.38–1.45]	43.6	<0.001	No	1.00	[0.40–0.84]	8.6	0.003
15–19	0.40	[0.23–0.68]			Yes	0.58			
20–24	1.00				Duration of operation; min				
≥25	3.10	[1.97–4.87]			0–60	1.00	[0.77–2.34]	39.9	<0.001
Use of adrenaline					61–90	1.35			
No	1.00		14.2	<0.001	>90	6.01	[3.25–11.12]		
Yes	0.54	[0.39–0.75]			Unknown	2.21	[1.30–3.77]		
Indication for CS – failure to progress in labour					Block adequate pre-operatively				
No	1.00		17.3	<0.001	No	6.04	[3.32–11.00]	30.9	<0.001
Yes	0.51	[0.37–0.70]			Yes	1.00			
Maternal weight; kg					Indication for CS–failed instrumental delivery				
≤59	0.69	[0.42–1.13]	13.8	0.008	No	1.00	[0.26–1.02]	4.1	0.044
60–66	1.00				Yes	0.51			
67–77	1.41	[0.92–2.18]			Baby's weight; g				
≥78	0.87	[0.54–1.40]			≤3000	1.31	[0.76–2.24]	8.4	0.039
Unknown	0.56	[0.30–1.06]			3001–3400	0.76			
					3401–3800	1.00	[0.39–0.98]		
					>3800	0.61			

general anaesthesia conversion rate in category 1 Caesarean section. The rate was 20% if cases of primary general anaesthesia with epidural analgesia that was not topped up are included as converted regionals, and 14% if they are not counted as such. Although this figure seems high, the alternative viewpoint is that by attempting to provide regional anaesthesia for 64% of women having category 1 Caesarean section, 56% have their operation with a regional and are spared the risks of general anaesthesia. Results from other units have also highlighted the difficulty of compliance with the RCA standard for general anaesthesia conversion in emergency Caesarean section. Garry & Davies had an 8% conversion rate in a single hospital (6% excluding cases with labour epidural analgesia that were not topped up) [9], and Jenkins & Khan found a 14.3% conversion rate in data from 14 hospitals (4.9% excluding those not topped up) [6].

One difference between elective and emergency Caesarean section which influences general anaesthesia conversion rates is the use of top-up epidurals in emergency cases. We had general anaesthesia conversion rates of 1.4% and 2.7% for spinal and CSE respectively, which are similar to those in a large US survey of 2.1% and 1.7% [7]. Our general anaesthesia conversion rate for epidural top-up anaesthesia was 8% (6.2% excluding those not topped up, usually because of urgency). This lies within a range of published figures from 5.3% (4.1% excluding not topped up) [8] to 13.5% (10.3%) [9]. The rate of pain during Caesarean section with spinal, CSE and epidural top-up anaesthesia was 3.5%, 13.4% and 13.7% respectively. It is important to note that we usually reserve CSE for cases with anaesthetic or surgical complexity.

One approach to improve anaesthetic reliability for the emergency Caesarean section with pre-existing epidural analgesia is the 'elective' use of spinal anaesthesia rather than topping up the epidural. Following this approach, Dresner et al. found a lower rate of pain during Caesarean section, although at the expense of a rate of high block needing respiratory support or intubation of as much as 1.5% [15]. In the current audit, the odds ratio for intra-operative failure of regional anaesthesia was 3.0 when a labour epidural had been used but not topped up compared to no prior epidural. There were no high blocks needing respiratory support or intubation in 174 cases of spinal after epidural during this audit. However, the data indicate that if a spinal is to be used in the presence of an epidural, it is preferable that the epidural has not been topped-up for surgery. In this audit, seven high sensory blocks, five with dyspnoea, occurred in 71 cases where a spinal was used after an unsuccessful epidural top-up using a mean spinal dose of 8 mg bupivacaine, compared to one high block in 68 cases where a spinal was used in the presence of an epidural that was not topped-up using a mean spinal dose of 12 mg bupivacaine. No high blocks were noted in 15 cases where an inadequate spinal was followed by a second spinal using a mean spinal dose of 9 mg bupivacaine.

For statistical analysis, we used two definitions of failure: pre-operative failure to achieve a satisfactory block and intra-operative failure leading to pain. The analysis confirms the importance of anaesthesia type and urgency. Duration of operation beyond 90 min increased the risk of intra-operative failure in the dataset of all those having regional anaesthesia, as well as in the spinal and epidural

top-up groups. We select CSE for cases with a perceived risk of prolonged surgery. Thus, CSE was used in 6% of cases with 0 or 1 previous Caesarean, but 22%, 59% and 80% of cases with 2, 3 or ≥ 4 Caesareans respectively.

An inadequate pre-operative block was associated with an increased risk of intra-operative failure. In many cases where surgery was started in the presence of an inadequate block, the reasons for this were unclear. However, 20% of these cases were started because of urgency or with a rising block. Block extent was assessed using cold sensation. The use of cold is more specific but less sensitive than touch [16]. This is borne out by the finding that 32% of general anaesthetic conversions occurred during the operation, mostly in the presence of an adequate pre-operative block. Although the use of touch to assess anaesthetic block for Caesarean section has been strongly recommended [16], there is a lack of data showing that it can reduce the rate of intra-operative pain or general anaesthetic conversion, or indeed overall general anaesthetic conversion [17].

The majority of spinals were inserted in the right lateral position. The sitting position was associated with a higher intra-operative failure rate using univariate analysis (5.1% versus right lateral 2.9%). Subdividing women who were sitting for spinal insertion into groups based on the indication for sitting was an attempt to separate clinically straightforward cases from those with obesity or predicted anatomical difficulty, which might have been confounding factors. Intra-operative failure occurred in 4.3% of cases where the spinal was inserted sitting because of the anaesthetist's preference. Other audit data has also suggested an increased chance of intra-operative pain if the spinal is inserted sitting [10]. However, using logistic regression analysis, insertion position was not found to be a significant factor affecting intra-operative failure.

It has been suggested that the use of spinal diamorphine does not affect the quality of block assessed with touch sensation [18], but reduces pain during surgery [19]. This audit found an increase in the chance of pre-operative failure to achieve a successful block to cold if a spinal opioid was not used, but no difference in intra-operative failure. Intra-operative pain was found in 3.5% of spinals with an opioid and 11.1% without, but as $< 1\%$ of spinals did not have a spinal opioid, the power to detect a real difference is lacking.

An association between nulliparity and increased pre-operative failure for spinals is unexplained. On the other hand, preterm delivery has been identified as a factor in failure to establish spinal block [20].

Several randomised studies comparing epidural top-up solutions have failed to find a clinically important difference between different local anaesthetics or with the

addition of fentanyl [11, 12, 21]. These studies assessed time of onset of surgical block but were too small to detect differences in failure rate. The use of different solutions in this audit was spread such as to allow reasonable numbers in each group for analysis (bupivacaine 58%, lidocaine 23%, lidocaine-bupivacaine mix 16%; adrenaline 46%; bicarbonate 26%; opioid 83%). Apart from adrenaline, these factors did not feature in the univariate or logistic regression analyses. Adrenaline, however, approximately halved the risk both of pre-operative and intra-operative failure. We have considered an epidural top-up volume of 20–24 ml to be 'standard' as anaesthetists usually draw up 20 ml of local anaesthetic and other adjuncts may then be included. Smaller volumes than 20 ml are associated with less risk of pre-operative failure possibly indicating adequacy of the existing analgesic block, whereas volumes of 25 ml or above are associated with diminishing returns in terms of success.

The most common indication for primary general anaesthesia was Caesarean section urgency associated with fetal distress, whether considering all cases, category 1 cases, or cases where epidural analgesia was used during labour but not topped up. Conversion of inadequate regional anaesthesia to general anaesthesia accounted for 35% of all women who had a general anaesthetic. This percentage is higher as the the primary general anaesthetic rate falls [4]. It is probably appropriate that 88% of general anaesthetic conversions in category 1 cases were pre-operative.

Hawkins et al. estimated that in the late 1980s, the risk of death from obstetric general anaesthesia was 17 times that of regional anaesthesia [22]. Shibli and Russell suggested a rate of serious life-threatening complications of 1 : 350 with general anaesthesia for Caesarean section and 1 : 2728 with regional anaesthesia [4]. The current study also indicates the relative safety of regional anaesthesia with a risk of serious incidents of 1 : 126 with general anaesthesia and 1 : 501 with regional anaesthesia. Critical incidents during general anaesthesia were mainly associated with conversion of regional anaesthesia rather than primary general anaesthesia.

Acknowledgements

I would like to thank all the anaesthetists who helped to collect the data, the anaesthetic audit staff and Hazel Taylor for statistical help.

References

- 1 Thomas J, Paranjothy S. *Royal College of Obstetricians and Gynaecologists Clinical Effectiveness Support Unit. The National Sentinel Caesarean Section Audit Report*. London:

- Royal College of Obstetricians and Gynaecologists Press, 2001.
- 2 Caesarean section: clinical guideline. *National Collaborating Centre for Women's and Children's Health; Commissioned by NICE*. London: Royal College of Obstetricians & Gynaecologists Press, 2004.
- 3 Colvin JR. *Raising the Standard. A Compendium of Audit Recipes for Continuous Quality Improvement in Anaesthesia*, 2nd edn. London: The Royal College of Anaesthetists, 2006.
- 4 Shibli KU, Russell IF. A survey of anaesthetic techniques used for caesarean section in the UK in 1997. *International Journal of Obstetric Anaesthesia* 2000; **9**: 160–7.
- 5 Lucas DN, Yentis SM, Kinsella SM, et al. Urgency of caesarean section: a new classification. *Journal of the Royal Society of Medicine* 2000; **93**: 346–50.
- 6 Jenkins JG, Khan MM. Anaesthesia for Caesarean section: a survey in a UK region from 1992 to 2002. *Anaesthesia* 2003; **58**: 1114–8.
- 7 Bloom SL, Spong CY, Weiner SJ, et al. Complications of anaesthesia for cesarean delivery. *Obstetrics & Gynecology* 2005; **106**: 281–7.
- 8 Riley ET, Papasin J. Epidural catheter function during labor predicts anesthetic efficacy for subsequent cesarean delivery. *International Journal of Obstetric Anaesthesia* 2002; **11**: 81–4.
- 9 Garry M, Davies S. Failure of regional blockade for caesarean section. *International Journal of Obstetric Anaesthesia* 2002; **11**: 9–12.
- 10 Habib AS, Dale M, Emerson C, Allsager C, Victoria A, Bourne TM. Reduction in pain during caesarean section following the addition of neuraxial opioids: a closed-loop audit. *International Journal of Obstetric Anaesthesia* 2002; **11**: S17.
- 11 Lucas DN, Ciccone GK, Yentis SM. Extending low-dose epidural analgesia for emergency caesarean section. A comparison of three solutions. *Anaesthesia* 1999; **54**: 1173–7.
- 12 Goring-Morris J, Russell IF. A randomised comparison of 0.5% bupivacaine with a lidocaine/epinephrine/fentanyl mixture for epidural top-up for emergency caesarean section after 'low dose' epidural for labour. *International Journal of Obstetric Anaesthesia* 2006; **15**: 109–14.
- 13 Inglis A, Daniel M, McGrady E. Maternal position during induction of spinal anaesthesia for Caesarean section. A comparison of right lateral and sitting positions. *Anaesthesia* 1995; **50**: 363–5.
- 14 Available at: http://www.oaa-anaes.ac.uk/assets/_managed/editor/File/PDF/Members/Documents/finalNOAD2004report.pdf (accessed 27 November 2007)
- 15 Dresner M, Brennan A, Freeman J. Six year audit of high regional blocks in obstetric anaesthesia. *International Journal of Obstetric Anaesthesia* 2003; **12**: S10.
- 16 Russell IF. At caesarean section under regional anaesthesia, it is essential to test sensory block with light touch before allowing surgery to start. *International Journal of Obstetric Anaesthesia* 2006; **15**: 294–7.
- 17 Dahlgren G. At caesarean section under regional anaesthesia, it is essential to test sensory block with light touch before allowing surgery to start. *International Journal of Obstetric Anaesthesia* 2006; **15**: 298–300.

- 18 Akerman N, Saxena S, Wilson R, Columb M, Lyons G. Effect of intrathecal diamorphine on block height during spinal anaesthesia for Caesarean section with bupivacaine. *British Journal of Anaesthesia* 2005; **94**: 843–7.
- 19 Saravanan S, Robinson APC, Quayoum Dar A, Columb MO, Lyons GR. Minimum dose of intrathecal diamorphine required to prevent intra-operative supplementation of spinal anaesthesia for Caesarean section. *British Journal of Anaesthesia* 2003; **91**: 368–72.
- 20 James KS, McGrady E, Patrick A. Combined spinal-extradural anaesthesia for preterm and term Caesarean section: is there a difference in local anaesthetic requirements. *British Journal of Anaesthesia* 1997; **78**: 498–501.
- 21 Malhotra S, Yentis SM. Extending low-dose epidural analgesia in labour for emergency Caesarean section – a comparison of levobupivacaine with or without fentanyl. *Anaesthesia* 2007; **62**: 667–71.
- 22 Hawkins JL, Koonin LM, Palmer SK, Gibbs CP. Anaesthesia-related deaths during obstetric delivery in the United States, 1979–1990. *Anesthesiology* 1997; **86**: 277–84.

Appendix

Flow diagram of conduct of regional anaesthesia from datasheet.

